EXP 9

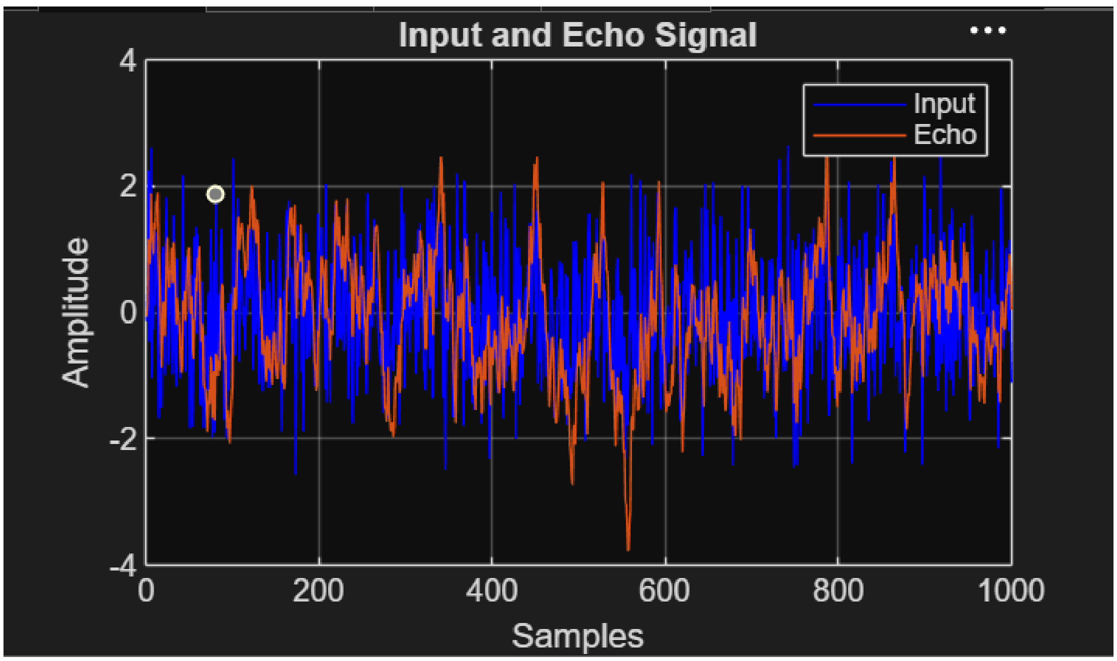
CODE:

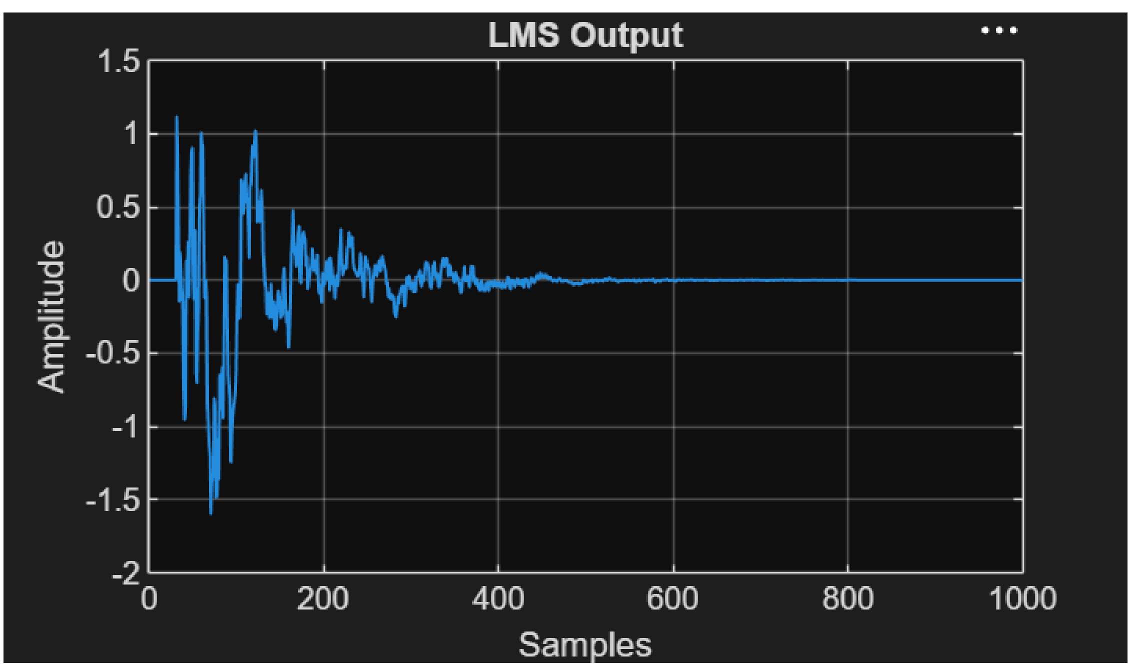
```
clc;
clear;
close all;
%% Parameters
N = 1000;                % Number of samples
L = 32;                  % Adaptive filter length
mu = 0.01;               % LMS Step size
mu_nlms = 0.8;           % NLMS Step size
delta = 1e-6;
%% Objective 1
% Generate Input Signal
x = randn(N,1);
% Echo Path (Unknown System)
h = exp(-0.12*(0:L-1))';
h = h/norm(h);
% Echo Signal
d = filter(h,1,x);
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%%%%%%%%%%
%% Objective 1
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%%%%%%%%%%
figure;
plot(x,'b');
hold on;
plot(d,'Color',[0.8500 0.3250 0.0980]);
grid on;
title('Input and Echo Signal');
legend('Input','Echo');
xlabel('Samples');
ylabel('Amplitude');
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%%%%%%%%%%
%% Objective 2 : LMS
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%%%%%%%%%%
w = zeros(L,1);
y_lms = zeros(N,1);
e_lms = zeros(N,1);
for n=L:N
```

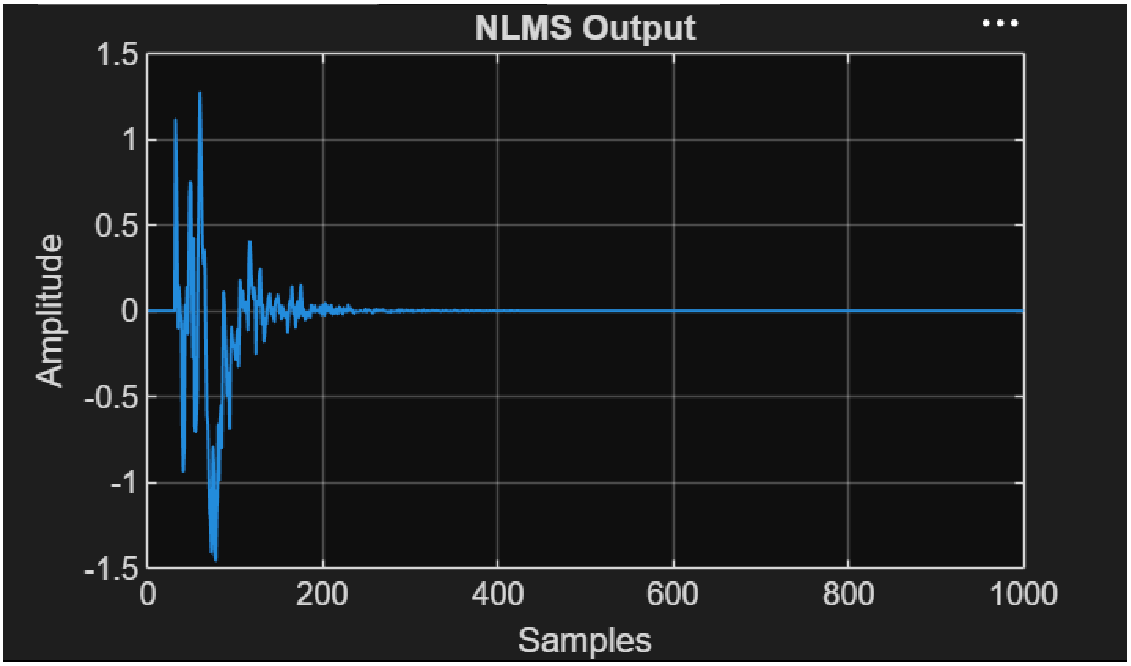
```
  xvec = x(n:-1:n-L+1);
  y_lms(n) = w'*xvec;
  e_lms(n) = d(n)-y_lms(n);
  w = w + mu*e_lms(n)*xvec;
end
figure;
plot(e_lms,'LineWidth',1);
grid on;
title('LMS Output');
xlabel('Samples');
ylabel('Amplitude');
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%%%%%%%%%%
%% Objective 3 : NLMS
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%%%%%%%%%%
w = zeros(L,1);
y_nlms = zeros(N,1);
e_nlms = zeros(N,1);
for n=L:N
  xvec = x(n:-1:n-L+1);
  y_nlms(n)=w'*xvec;
  e_nlms(n)=d(n)-y_nlms(n);
  w=w+(mu_nlms/(delta+xvec'*xvec))*e_nlms(n)*xvec;
end
figure;
plot(e_nlms,'LineWidth',1);
grid on;
title('NLMS Output');
xlabel('Samples');
ylabel('Amplitude');
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%%%%%%%%%%
%% Objective 4 : MSE Comparison
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%%%%%%%%%%
MSE_LMS=zeros(N,1);
MSE_NLMS=zeros(N,1);
for k=1:N
  MSE_LMS(k)=mean(e_lms(1:k).^2);
  MSE_NLMS(k)=mean(e_nlms(1:k).^2);
end
figure;
semilogy(MSE_LMS,'r','LineWidth',1.2);
```

```
hold on;
semilogy(MSE_NLMS,'b','LineWidth',1.2);
grid on;
legend('LMS','NLMS');
title('MSE Comparison');
xlabel('Iterations');
ylabel('Mean Square Error');
```

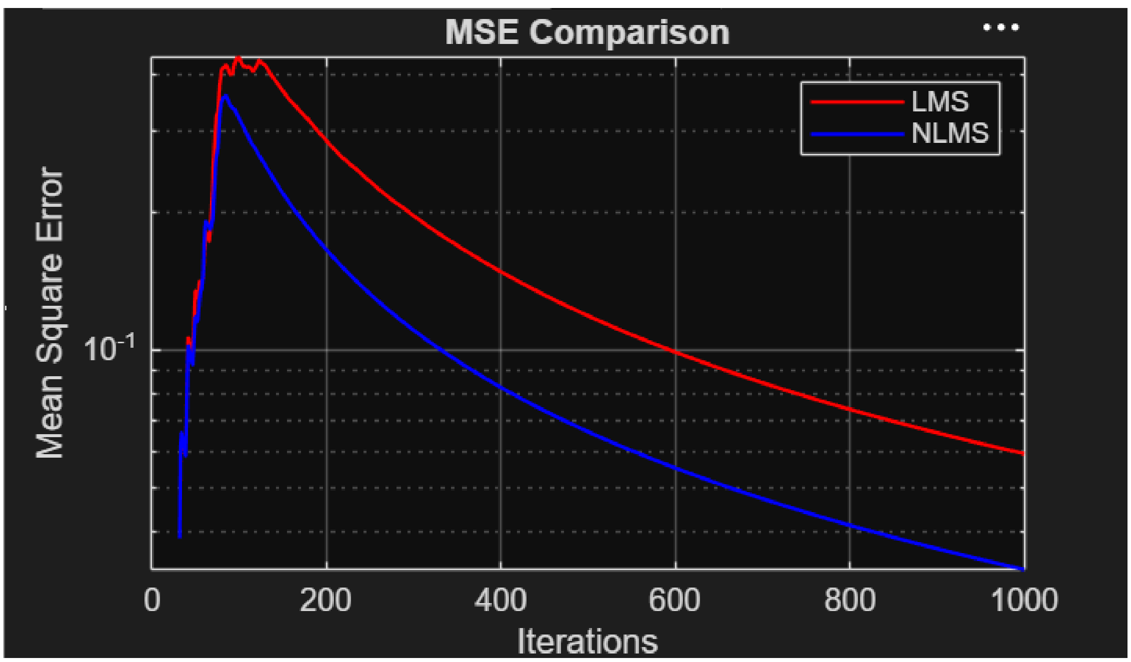
OUTPUT:







NLMS Output
Amplitude
Samples
1.5
1
0.5
0
-0.5
-1
-1.5
0
200
400
600
800
1000



MSE Comparison
LMS
NLMS
Mean Square Error
Iterations
10^{-1}
0
200
400
600
800
1000